

Philip Hargreaves

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Summary

MSc Computer Science at UCL with two years' nuclear R&D experience at Rolls-Royce. Comfortable across full-stack web (React/Next.js), backend (Go, Node.js, PHP), and concurrent/parallel systems (C++ , Go).

Education

University College London	Sept 2025 – Sept 2026
MSc Computer Science	Expected Distinction

- **Modules:** Data Structures and Algorithms, Computer Architecture, Operating Systems, Research Computing with C++ , Object-Oriented Programming, Machine Learning, Database Fundamentals

University of Nottingham	Sept 2018 – Sept 2022
MEng Mechanical Engineering	First Class Honours

Experience

Mechanical Design Engineer , Rolls-Royce	May 2023 – Sept 2025
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- Developed a Python-based design toolkit using Model-View-Presenter architecture and Tkinter GUI, integrating NumPy, Pandas, and Matplotlib, reducing preliminary vessel sizing time by 30%
- Achieved a 40% life extension for a critical nuclear component by developing refined brittle fracture methodologies, securing ONR regulatory approval and >£50k in recurring operational savings
- Automated FEA post-processing using Python, decreasing manual effort and ensuring data consistency
- Validated sub-contractor deliverables as technical reviewer, verifying engineering calculations and models to ensure 100% compliance with nuclear safety case requirements

Projects

CS Events Platform , UCL Department of Computer Science	React, Next.js, TypeScript, Jest, Azure
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- Delivered Next.js/React events platform built on PostgreSQL/Prisma using feature-driven architecture with full unit/integration/E2E test coverage, piloted by the department to replace current fragmented workflows
- Built conversational AI assistant auto-filling event forms from messages or PDFs/images via Azure GPT-4.1
- Engineered Microsoft Graph cron pipeline transforming Outlook ICS invites into draft events, with single-use, DB-hashed JWT edit links emailed to external organisers and consumed atomically in Prisma

Full-Stack Auction Platform	PHP 8.4, MySQL, JavaScript, Apache, Docker
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- Created custom RESTful MVC framework in vanilla PHP 8.4 routing 34 endpoints via dependency injection and repository pattern, with a normalised 14-table MySQL schema supporting recursive category filtering
- Developed transaction-safe bidding engine with outbid detection, reserve-price checks and winner resolution
- Designed weighted recommendation system using collaborative filtering, in-memory caching and SQL CTEs

Agent-Based Ecological Simulation	C++ 17, OpenMP, MPI, CMake, Catch2
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- Built evolutionary predator-prey simulation with behavioural state machine, genetic trait inheritance and natural population dynamics, parallelised across 4 distributed processes and multi-core OpenMP threads
- Engineered cross-process reproducibility via seed broadcast, with Catch2-verified behaviour and genetics

Feed Ingestion Service	Go, PostgreSQL, SQLC, Goose
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- Built Go service that ingests RSS feeds into PostgreSQL, normalises entries and serves personalised streams
- Implemented concurrent ingestion using worker pools, batch scheduling and domain-level rate limiting
- Developed a process supervisor with exponential backoff to auto-recover from network or panic failures

Skills

Programming Languages: Python, Go, C, C++ , TypeScript, JavaScript, PHP, Java, SQL, Bash

Frameworks / Libraries: React, Next.js, Node.js, Prisma, OpenMP, MPI, Jest, Playwright, Catch2, Pytest

Tools / Cloud: Git, Docker, Linux, AWS (Certified Cloud Practitioner), Azure, PostgreSQL, MySQL, CMake